# The width of an $n$-good array

golden conjugates, an unreachable integer, and how far you must miss it

**Problem.** For an integer $n \geq 4$, call a real tuple $(x_1, \dots, x_n)$ an $n$*-good array* if

$$\sum_{i=1}^{n} x_i = n, \qquad \sum_{i=1}^{n} x_i^2 = 2n, \qquad \sum_{i=1}^{n} x_i^3 = 3n,$$

and let $\operatorname{width}(x) = \max_i x_i - \min_i x_i$.

(1) Find the largest real $C$ with $\operatorname{width}(x) \geq C$ for every $n \geq 4$ and every $n$-good array.

(2) For that $C$, prove that some $\lambda > 0$ satisfies $\operatorname{width}(x) > C + \lambda n^{-3/2}$ for every $n \geq 4$ and every $n$-good array.

$$\boxed{C = \sqrt{5}; \qquad \operatorname{width}(x) > \sqrt{5} + \frac{\lambda}{n^{3/2}} \text{ holds with } \lambda = \tfrac{1}{8}.}$$

Sections 1–9 prove both statements. Sections 10–13 are a separate, numerical study of how far the exponent $\frac{3}{2}$ and the constant $\lambda$ can be pushed; the results there are stated as conjectures, with the evidence given.

**Overview.** The three conditions are not independent constraints to be juggled: they are a disguised statement about the golden ratio. Let

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad a = \frac{1-\sqrt{5}}{2},$$

the two roots of $t^2 = t + 1$, so that also $t^3 = 2t + 1$ for $t \in \{a, \varphi\}$. If *every* $x_i$ lies in $\{a, \varphi\}$, then

$$\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i + n, \qquad \sum_{i=1}^{n} x_i^3 = 2\sum_{i=1}^{n} x_i + n,$$

so the first condition $\sum x_i = n$ forces the other two automatically. Such an array is $n$-good, has width $\varphi - a = \sqrt{5}$, and is the only possible way to achieve width $\sqrt{5}$.

But it cannot exist. If $k$ entries equal $\varphi$ and $n-k$ equal $a$, then $\sum x_i = n$ reads $k\sqrt{5} = n(1-a)$, i.e.

$$k = np, \qquad p := \frac{1-a}{\sqrt{5}} = \frac{5+\sqrt{5}}{10} = 0.7236\dots,$$

and $np$ is irrational, never an integer. The ideal configuration is arithmetically forbidden, so width $> \sqrt{5}$ strictly. It can still be approached, because $np$ can be made arbitrarily close to an integer. That is part (1).

Part (2) asks how badly the near-misses must fail, and the answer is governed by how close $np$ can be to $\mathbb{Z}$. Since $p$ is a quadratic irrational, $\|np\|_{\mathbb{Z}} \geq \frac{2}{5n}$; and an array of width $\sqrt{5}(1+\eta)$ can be shown to force $\|np\|_{\mathbb{Z}} \ll n^2\eta^2$. Comparing the two gives $\eta \gg n^{-3/2}$.

Sections 10–13 then ask what the narrowest $n$-good array actually looks like. Numerically it has three distinct values, and its width follows a parametric law $M_n = \sqrt{5} + \sqrt{5}\,\Phi(\{np\})/n + O(n^{-2})$ rather than a single power of $n$: the excess is of order $\frac{1}{n}$ for most $n$ and drops to order $n^{-3/2}$ only along the odd-index Lucas numbers, where $\{np\}$ is closest to 1. None of this is proved here, and Section 12 states precisely what would be needed.

# 1. Normalisation: send $\{a,\varphi\}$ to $\{0,1\}$

Put

$$x_i = a + \sqrt{5}\,u_i, \qquad \text{equivalently} \qquad u_i = \frac{x_i - a}{\sqrt{5}}, \qquad \operatorname{width}(x) = \sqrt{5}\,\operatorname{width}(u), \tag{1}$$

so that $x_i = a$ corresponds to $u_i = 0$ and $x_i = \varphi$ to $u_i = 1$.

**Lemma 1.** With $p = \frac{5+\sqrt{5}}{10}$, the tuple $x$ is $n$-good if and only if

$$\boxed{\sum_{i=1}^n u_i = \sum_{i=1}^n u_i^2 = \sum_{i=1}^n u_i^3 = np.} \tag{2}$$

*Proof.* Use $a^2 = a+1$ and $a^3 = 2a+1$. Expanding (1),

$$\sum_{i=1}^n x_i = na + \sqrt{5}\sum_{i=1}^n u_i,$$

so $\sum x_i = n$ is $\sum u_i = \frac{n(1-a)}{\sqrt{5}} = np$. Next,

$$\sum_{i=1}^n x_i^2 = na^2 + 2a\sqrt{5}\sum_{i=1}^n u_i + 5\sum_{i=1}^n u_i^2,$$

and substituting $a^2 = a+1$ and $\sum u_i = np = \frac{n(1-a)}{\sqrt{5}}$,

$$5\sum_{i=1}^n u_i^2 = 2n - n(a+1) - 2an(1-a) = n(1-a)(1-2a) = n(1-a)\sqrt{5},$$

since $1-2a = \sqrt{5}$; hence $\sum u_i^2 = \frac{n(1-a)}{\sqrt{5}} = np$. Similarly

$$\sum_{i=1}^n x_i^3 = na^3 + 3a^2\sqrt{5}\sum_{i=1}^n u_i + 15a\sum_{i=1}^n u_i^2 + 5\sqrt{5}\sum_{i=1}^n u_i^3,$$

and using $a^3 = 2a+1$, $a^2 = a+1$ and $3a(1+\sqrt{5}) = -6$,

$$5\sqrt{5}\sum_{i=1}^n u_i^3 = n(1-a)\left[2 - 3a^2 - 3\sqrt{5}\,a\right] = 5n(1-a),$$

so $\sum u_i^3 = np$. Every step is reversible. ☐

The three inhomogeneous conditions have become one homogeneous statement: *the first three power sums of $u$ coincide.* The specific value $np$ they share is the entire arithmetic content of the problem.

# 2. The test polynomial

Let

$$q(t) = t(1-t).$$

By (2),

$$\sum_{i=1}^n q(u_i) = \sum_{i=1}^n u_i - \sum_{i=1}^n u_i^2 = 0, \qquad \sum_{i=1}^n u_i\,q(u_i) = \sum_{i=1}^n u_i^2 - \sum_{i=1}^n u_i^3 = 0. \tag{3}$$

So $q$ has mean zero against both the counting measure and the measure weighted by $u$. Since $q \geq 0$ exactly on $[0, 1]$, the first identity in (3) already says that the $u_i$ cannot all lie strictly inside $[0, 1]$ unless they sit at the endpoints. The two identities together pin both ends.

**Lemma 2.** Let $m = \min_i u_i$ and $M = \max_i u_i$. Then $M \geq 1$ and $m \leq 0$.

*Proof.* Suppose $M < 1$. Then $1 - u_i > 0$ for all $i$, so by (3)

$$0 = \sum_{i=1}^{n} u_i\, q(u_i) = \sum_{i=1}^{n} u_i^2(1 - u_i)$$

is a sum of non-negative terms; hence $u_i^2(1 - u_i) = 0$, i.e. $u_i = 0$ for every $i$. But then $\sum u_i = 0 \neq np$.

Suppose $m > 0$. Subtracting the two identities in (3),

$$0 = \sum_{i=1}^{n} (u_i - 1)q(u_i) = -\sum_{i=1}^{n} u_i(u_i - 1)^2,$$

again a sum of non-negative terms since $u_i > 0$; hence $u_i = 1$ for every $i$, and $\sum u_i = n \neq np$ because $p \neq 1$. □

## 3. Part (1), lower bound: $\mathrm{width}(x) > \sqrt{5}$

By Lemma 2,

$$\mathrm{width}(u) = M - m \geq 1 - 0 = 1.$$

Suppose $\mathrm{width}(u) = 1$. Then $m = 0$ and $M = 1$ (Lemma 2 forces $m \leq 0 \leq 1 \leq M$, and $M - m = 1$ leaves no slack), so $0 \leq u_i \leq 1$ for all $i$. On $[0, 1]$ we have $q \geq 0$, and $\sum q(u_i) = 0$ by (3), so $q(u_i) = 0$ for every $i$, i.e.

$$u_i \in \{0, 1\} \quad \text{for all } i.$$

Then $\sum u_i$ is a non-negative integer. But (2) gives

$$\sum_{i=1}^{n} u_i = np = n \cdot \frac{5 + \sqrt{5}}{10} \notin \mathbb{Z},$$

since $\sqrt{5}$ is irrational and $n \geq 1$. Contradiction. Therefore

$$\boxed{\mathrm{width}(u) > 1, \qquad \mathrm{width}(x) = \sqrt{5}\,\mathrm{width}(u) > \sqrt{5}.} \tag{4}$$

Unwinding the normalisation, the excluded configuration is exactly the one described in the overview: all entries in $\{a, \varphi\}$, which would be $n$-good but requires $np \in \mathbb{Z}$.

## 4. Part (1), sharpness: a three-point patch

It remains to show that $\sqrt{5}$ is approached, so that no larger constant works. The idea is to keep most entries at 0 and 1 and absorb the fractional part of $np$ into three extra entries.

**Lemma 3.** For every $R \in (1, 2)$ there exist three distinct reals $y_1, y_2, y_3$ with

$$y_1 + y_2 + y_3 = y_1^2 + y_2^2 + y_3^2 = y_1^3 + y_2^3 + y_3^3 = R. \tag{5}$$

*Proof.* Prescribe the power sums $s_1 = s_2 = s_3 = R$ and read off the elementary symmetric functions from Newton's identities:

$$e_1 = R, \qquad e_2 = \frac{s_1 e_1 - s_2}{2} = \frac{R^2 - R}{2}, \qquad e_3 = \frac{e_2 s_1 - e_1 s_2 + s_3}{3} = \frac{R(R-1)(R-2)}{6}.$$

The corresponding monic cubic is

$$F_R(y) = y^3 - Ry^2 + \frac{R^2 - R}{2}\, y - \frac{R(R-1)(R-2)}{6}, \tag{6}$$

whose discriminant is

$$\Delta(F_R) = \frac{R^2(3-R)^2(2-R)(R-1)}{6}. \tag{7}$$

For $1 < R < 2$ all four factors are positive, so $\Delta > 0$ and $F_R$ has three distinct real roots; by construction they satisfy (5). $\square$

(Identity (7) is the ordinary cubic discriminant $18e_1e_2e_3 - 4e_1^3e_3 + e_1^2e_2^2 - 4e_2^3 - 27e_3^2$ after simplification; it was verified symbolically and numerically.)

**The construction.** Since $p$ is irrational, $\{np\}_{n\geq 1}$ is equidistributed mod 1; more concretely, if $p_k/q_k$ are the continued-fraction convergents of $p$, then for those $k$ with $q_kp - p_k > 0$ one has

$$0 < \operatorname{frac}(q_kp) = q_kp - p_k < \frac{1}{q_k},$$

and these occur for infinitely many $k$. So choose $n_j \to \infty$ with

$$r_j := \operatorname{frac}(n_jp) \longrightarrow 0^+,$$

and set

$$K_j = \lfloor n_jp \rfloor - 1, \qquad R_j = 1 + r_j \in (1,2), \qquad \text{so that} \qquad K_j + R_j = n_jp. \tag{8}$$

For large $j$ both $K_j \geq 0$ and $n_j - K_j - 3 \geq 0$ (as $K_j \approx 0.7236\, n_j$). Take the array

$$u^{(j)} = \big(\underbrace{1,\dots,1}_{K_j},\ \underbrace{0,\dots,0}_{n_j-K_j-3},\ y_1, y_2, y_3\big), \qquad y_i \text{ the roots of } F_{R_j}.$$

Then for $k = 1,2,3$,

$$\sum_i \big(u_i^{(j)}\big)^k = K_j \cdot 1 + 0 + R_j = n_jp,$$

so by Lemma 1 the corresponding $x^{(j)}$ is $n_j$-good.

As $R_j \to 1^+$, the cubic (6) tends to $y^3 - y^2 = y^2(y-1)$, so the root multiset converges: $\{y_1,y_2,y_3\} \to \{0,0,1\}$. Hence $\operatorname{width}(u^{(j)}) \to 1$ and

$$\operatorname{width}(x^{(j)}) = \sqrt{5}\ \operatorname{width}(u^{(j)}) \longrightarrow \sqrt{5}.$$

Combined with (4),

$$\boxed{\inf \operatorname{width}(x) = \sqrt{5}, \qquad C = \sqrt{5}.}$$

This settles part (1). The patched array is not the narrowest $n$-good array, though: it spends three entries on the cubic, and Section 10 exhibits a minimiser using only one interior value, whose excess over $\sqrt{5}$ is smaller by a factor of order $n$.

| $n$ | $\operatorname{frac}(np)$ | $\operatorname{width}(u)$ | $\eta = \operatorname{width}(u) - 1$ | $\operatorname{width}(x)$ |
|---|---|---|---|---|
| 14 | $1.30 \cdot 10^{-1}$ | 1.1448171 | $1.45 \cdot 10^{-1}$ | 2.5598889 |
| 123 | $3.64 \cdot 10^{-3}$ | 1.0246280 | $2.46 \cdot 10^{-2}$ | 2.2911378 |
| 1165 | $1.92 \cdot 10^{-3}$ | 1.0178901 | $1.79 \cdot 10^{-2}$ | 2.2760715 |
| 15127 | $2.96 \cdot 10^{-5}$ | 1.0022198 | $2.22 \cdot 10^{-3}$ | 2.2410315 |
| 103682 | $4.31 \cdot 10^{-6}$ | 1.0008479 | $8.48 \cdot 10^{-4}$ | 2.2379639 |
| $\infty$ | 0 | 1 | 0 | $\sqrt{5} = 2.2360680$ |

The three power sums of each row were verified to 25 decimal places.

## 5. Part (2), setup

Write

$$\operatorname{width}(u) = M - m = 1 + \eta, \qquad \eta > 0 \text{ by (4)}.$$

If $\eta \geq 1$ then $\operatorname{width}(x) \geq 2\sqrt{5} > \sqrt{5} + \lambda n^{-3/2}$ for any $\lambda \leq 1$, and there is nothing to prove; so assume $0 < \eta < 1$ from now on.

By Lemma 2, $m \leq 0$ and $M \geq 1$; together with $M - m = 1 + \eta$ this gives $m \geq M - 1 - \eta \geq -\eta$ and $M \leq m + 1 + \eta \leq 1 + \eta$, so

$$\boxed{-\eta \leq u_i \leq 1 + \eta \quad \text{for all } i.} \tag{9}$$

Every entry is trapped in a window of width $1 + 2\eta$ around $[0, 1]$. Write $\|t\|_{\mathbb{Z}} = \min_{K \in \mathbb{Z}} |t - K|$ and $d_i = \operatorname{dist}(u_i, \{0, 1\})$.

Sections 6 to 8 bound $\sum d_i$ by $n\eta$, upgrade that to a bound on $\sum d_i^2$, and convert it into an upper bound for $\|np\|_{\mathbb{Z}}$. Section 8 contrasts this with a lower bound coming from the irrationality of $\sqrt{5}$.

## 6. The entries are close to $\{0, 1\}$ in $\ell^1$

On $[0, 1]$, $q(t) = t(1 - t) = \min(t, 1 - t) \cdot \max(t, 1 - t) \geq \frac{1}{2} d(t)$, where $d(t) = \operatorname{dist}(t, \{0, 1\})$; that is,

$$d(t) \leq 2q(t), \qquad 0 \leq t \leq 1. \tag{10}$$

On the two flanks $t \in [-\eta, 0] \cup [1, 1 + \eta]$,

$$q(t) \geq -\eta(1 + \eta) \geq -2\eta, \qquad d(t) \leq \eta. \tag{11}$$

By (3) and (11), splitting according to whether $u_i \in [0, 1]$,

$$\sum_{u_i \in [0,1]} q(u_i) = - \sum_{u_i \notin [0,1]} q(u_i) \leq 2n\eta,$$

and therefore, by (10) and (11) again,

$$\sum_{i=1}^{n} d_i = \sum_{u_i \in [0,1]} d_i + \sum_{u_i \notin [0,1]} d_i \leq 2 \sum_{u_i \in [0,1]} q(u_i) + n\eta \leq 5n\eta. \tag{12}$$

Since the $d_i$ are non-negative, $(\sum d_i)^2 = \sum d_i^2 + (\text{cross terms}) \geq \sum d_i^2$, so

$$\sum_{i=1}^{n} d_i^2 \leq \Big( \sum_{i=1}^{n} d_i \Big)^2 \leq 25\, n^2 \eta^2. \tag{13}$$

## 7. A doubly tangent cubic converts $\ell^2$ into an integer

The bound (13) is quadratic in the distances, so it must be paired with a quantity that is itself quadratically small near $\{0, 1\}$. Take

$$B(t) = 3t^2 - 2t^3.$$

It satisfies $B(0) = 0$, $B(1) = 1$, and $B'(t) = 6t(1 - t)$ vanishes at both endpoints; equivalently

$$B(t) = t^2(3 - 2t), \qquad 1 - B(t) = (1 - t)^2(1 + 2t),$$

which exhibits the double contact directly. Consequently, for $-1 \leq t \leq 2$,

$$\operatorname{dist}\big(B(t), \{0, 1\}\big) \leq 5 \operatorname{dist}(t, \{0, 1\})^2, \tag{14}$$

since $|B(t)| \le 5t^2$ for $t \le \frac{1}{2}$ and $|1 - B(t)| \le 5(1-t)^2$ for $t \ge \frac{1}{2}$ on that range.

The point of choosing $B$ among all doubly tangent cubics is that its sum is computable from (2):

$$\sum_{i=1}^{n} B(u_i) = 3\sum_{i=1}^{n} u_i^2 - 2\sum_{i=1}^{n} u_i^3 = 3np - 2np = np. \tag{15}$$

Now let $b_i \in \{0,1\}$ be a nearest Boolean value to $B(u_i)$, and put $K = \sum_i b_i \in \mathbb{Z}$. By (9) we have $u_i \in [-1, 2]$, so (14) applies, and with (15), (13),

$$\|np\|_{\mathbb{Z}} \le \Big|np - K\Big| = \Big|\sum_{i=1}^{n} B(u_i) - \sum_{i=1}^{n} b_i\Big| \le \sum_{i=1}^{n} \mathrm{dist}\big(B(u_i), \{0,1\}\big) \le 5\sum_{i=1}^{n} d_i^2 \le 125\, n^2\eta^2. \tag{16}$$

# 8. $np$ cannot be too close to an integer

**Lemma 4.** For every integer $n \ge 4$,

$$\|np\|_{\mathbb{Z}} \ge \frac{2}{5\,n}, \qquad p = \frac{5+\sqrt{5}}{10}. \tag{17}$$

*Proof.* Let $K \in \mathbb{Z}$ and put $m = 2K - n \in \mathbb{Z}$, so that $m \equiv n \pmod 2$ and

$$10\,|np - K| = |5n + n\sqrt{5} - 10K| = |n\sqrt{5} - 5m|.$$

If $|n\sqrt{5} - 5m| \ge 1$ then $|np - K| \ge \frac{1}{10} \ge \frac{2}{5n}$ for $n \ge 4$. Otherwise $|n\sqrt{5} - 5m| < 1$, which forces $5m > n\sqrt{5} - 1 > 0$ and

$$n\sqrt{5} + 5m < 2\sqrt{5}\,n + 1 \le 5n \qquad (n \ge 2).$$

Now use the parity of $m$. If $n$ and $m$ are both even then $4 \mid n^2$ and $4 \mid m^2$; if both are odd then $n^2 \equiv m^2 \equiv 1 \pmod 8$, so $n^2 - 5m^2 \equiv -4 \equiv 0 \pmod 4$. In either case $4 \mid n^2 - 5m^2$, and $n^2 - 5m^2 \ne 0$ because $\sqrt{5}$ is irrational. Hence $|n^2 - 5m^2| \ge 4$ and

$$|n\sqrt{5} - 5m| = \frac{5\,|n^2 - 5m^2|}{n\sqrt{5} + 5m} \ge \frac{5 \cdot 4}{5n} = \frac{4}{n},$$

so $|np - K| \ge \frac{2}{5n}$. Taking the minimum over $K$ gives (17). □

The divisibility $4 \mid n^2 - 5m^2$ is what upgrades the naive $|n^2 - 5m^2| \ge 1$ to $\ge 4$, and it costs nothing: it is only the observation that $m$ and $n$ have the same parity by construction.

Bound (17) is close to best possible. Numerically $\min_{4 \le n \le 3\cdot 10^5} n\,\|np\|_{\mathbb{Z}} = 0.4222912$, attained at $n = 4$, against the constant $\frac{2}{5} = 0.4$ of (17); Section 12 shows that the true asymptotic constant is $\frac{1}{\sqrt{5}} = 0.4472136$. Note that $\|np\|_{\mathbb{Z}} \asymp n^{-1}$ is *false* as a statement about all $n$: for most $n$ the quantity $\|np\|_{\mathbb{Z}}$ is of order one, and the content of (17) is the lower bound alone.

# 9. Part (2), conclusion

Combining (16) and (17), for $0 < \eta < 1$ and $n \ge 4$,

$$\frac{2}{5n} \le \|np\|_{\mathbb{Z}} \le 125\,n^2\eta^2 \qquad \Longrightarrow \qquad \eta^2 \ge \frac{2}{625\,n^3},$$

that is,

$$\boxed{\eta \ge \frac{\sqrt{2}}{25\,n^{3/2}} = \frac{0.0565685\ldots}{n^{3/2}}.} \tag{18}$$

(The case $\eta \geq 1$ was settled in Section 5, where the bound is trivial.) Hence

$$\mathrm{width}(x) = \sqrt{5}\,\mathrm{width}(u) = \sqrt{5}(1+\eta) \geq \sqrt{5} + \frac{\sqrt{10}}{25\,n^{3/2}} = \sqrt{5} + \frac{0.1264911\ldots}{n^{3/2}},$$

and therefore, with the explicit choice $\lambda = \frac{1}{8} = 0.125$,

$$\boxed{\mathrm{width}(x) > \sqrt{5} + \frac{\lambda}{n^{3/2}} \qquad \text{for every } n \geq 4 \text{ and every } n\text{-good array.}}$$

The chain (12)–(18) is quantitative at every step, so $\lambda$ comes out explicitly rather than from a compactness or contradiction argument.

# 10. A candidate minimiser

Sections 10–13 are not part of the proof. They describe the narrowest $n$-good array that numerical search finds and work out its asymptotics. The formulas are supported by computation and by a consistent asymptotic expansion, but global optimality is not proved; Section 12 states which steps are missing.

Write

$$M_n = \inf\bigl\{\mathrm{width}(x) : x \text{ is } n\text{-good}\bigr\}, \qquad W_n = \frac{M_n}{\sqrt{5}} = \inf \mathrm{width}(u),$$

so that part (2) asserts $M_n > \sqrt{5} + \lambda n^{-3/2}$.

The configuration found by numerical minimisation takes three distinct values, with the two extreme ones carrying almost all the mass.

**The ansatz.** Fix $n$ and put

$$k = \lfloor np \rfloor, \qquad \ell = n - k - 1, \qquad r = \{np\},$$

and consider arrays of the form

$$u = \bigl(\underbrace{v_-, \ldots, v_-}_{\ell},\ w,\ \underbrace{v_+, \ldots, v_+}_{k}\bigr), \qquad v_- < w < v_+. \tag{19}$$

The three moment equations (2) become three equations in $v_-, w, v_+$,

$$\ell v_-^j + w^j + k\,v_+^j = np, \qquad j = 1, 2, 3, \tag{20}$$

so no free parameter remains. By Lemma 2 the extreme values satisfy $v_- \leq 0$ and $v_+ \geq 1$, and by Section 3 at least one inequality is strict; numerically both are.

**Reduction to one scalar equation.** System (20) is easier to handle after eliminating two unknowns. For fixed $w$, the first two equations determine $v_\pm$ uniquely: with $T = n-1$ and $s = np$,

$$\mu(w) = \frac{s-w}{T}, \qquad D(w) = s - w^2 - \frac{(s-w)^2}{T}, \qquad \delta(w) = \sqrt{\frac{T\,D(w)}{k\ell}},$$

$$v_-(w) = \mu(w) - \frac{k}{T}\delta(w), \qquad v_+(w) = \mu(w) + \frac{\ell}{T}\delta(w), \tag{21}$$

and the third equation becomes a single scalar condition

$$G_n(w) = \ell\,v_-(w)^3 + w^3 + k\,v_+(w)^3 - s = 0. \tag{22}$$

For every $n$ checked, $4 \le n \le 10^4$, one finds $G_n(0) > 0$, $G_n(1) < 0$, and exactly one root in between, giving an ordered solution $v_- < 0 < w < 1 < v_+$. Proving these two sign conditions and the uniqueness of the root would establish existence and uniqueness of (19); that is not done here.

**Asymptotics.** Write $v_- = -A/n$ and $v_+ = 1 + C/n$ and expand (20) in $1/n$, using $\ell = (1-p)n + O(1)$ and $k = pn + O(1)$. To leading order,

$$p\,C = w^2(1-w) + O(n^{-1}), \qquad (1-p)\,A = w(1-w)^2 + O(n^{-1}), \tag{23}$$

and substituting back leaves a single condition on $w$:

$$\boxed{3w^2 - 2w^3 = \{np\} + O(n^{-1}).} \tag{24}$$

The error term in (24) is not negligible at small $n$. At $n = 4$ the actual root is $w_4 = 0.8155117\ldots$, whereas the solution of $3w^2 - 2w^3 = \{4p\}$ is $0.7985\ldots$; the two agree only to first order.

The cubic on the left of (24) is exactly the doubly tangent $B$ of Section 7, and the coincidence has a reason. Identity (15) says $\sum_i B(u_i) = np$ for *every* good array; evaluated on (19), $B(v_-) = O(n^{-2})$ and $1 - B(v_+) = O(n^{-2})$, so

$$\ell B(v_-) + B(w) + kB(v_+) = B(w) + k + O(n^{-1}) = np \quad\Longrightarrow\quad B(w) = \{np\} + O(n^{-1}).$$

The function used to prove the lower bound is the same one that locates the middle atom. Section 12 shows that this is why the bound of Section 9 is sharp in order.

**Inverting $B$.** On $[0,1]$ the map $B$ is an increasing bijection onto $[0,1]$, and its inverse has a closed form:

$$\boxed{\sigma(r) = \frac{1}{2} - \sin\left(\frac{1}{3}\arcsin(1-2r)\right), \qquad B(\sigma(r)) = r, \quad r \in [0,1].} \tag{25}$$

(Endpoints: $r = 0$ gives $\frac{1}{2} - \sin\frac{\pi}{6} = 0$, $r = 1$ gives $\frac{1}{2} + \sin\frac{\pi}{6} = 1$, and $r = \frac{1}{2}$ gives $\frac{1}{2}$.) Formula (25) is the trigonometric solution of $2\sigma^3 - 3\sigma^2 + r = 0$, which has three real roots for $r \in (0,1)$; (25) selects the one in $[0,1]$.

# 11. The parametric law

Substituting (23) and (24) into $\mathrm{width}(u) = v_+ - v_- = 1 + \frac{A+C}{n}$ gives the following description, again asymptotic and again unproved.

**Conjecture.** With $\sigma = \sigma(\{np\})$ from (25) and

$$\boxed{\Phi(r) = \sigma(1-\sigma)\left(\frac{\sigma}{p} + \frac{1-\sigma}{1-p}\right), \qquad 1 - p = \frac{5-\sqrt{5}}{10},} \tag{26}$$

the minimal width satisfies

$$\boxed{W_n = 1 + \frac{\Phi(\{np\})}{n} + O\left(\frac{1}{n^2}\right), \qquad M_n = \sqrt{5} + \frac{\sqrt{5}\,\Phi(\{np\})}{n} + O\left(\frac{1}{n^2}\right).} \tag{27}$$

Two caveats on (27). It presumes that (19) is optimal, which is not proved. And the remainder $O(n^{-2})$ is claimed uniformly, whereas the expansion leading to it is not uniform: $B'(\sigma) \to 0$ as $\{np\} \to 0$ or 1, so the implicit-function step degenerates precisely on the subsequences that matter in Section 12. Numerically the remainder does behave like $n^{-2}$ there, but the derivation above does not establish it.

Granting (27), the answer is not a single power of $n$: it is a fixed profile $\Phi$ sampled at the equidistributed points $\{np\}$ and scaled by $\frac{1}{n}$. For most $n$ the excess is of order $\frac{1}{n}$; smaller powers appear only when $\{np\}$ approaches an endpoint.

Some values of $\Phi$. Since

$$p(1-p) = \frac{5+\sqrt{5}}{10} \cdot \frac{5-\sqrt{5}}{10} = \frac{20}{100} = \frac{1}{5},$$

the midpoint value is exact:

$$\Phi\left(\tfrac{1}{2}\right) = \frac{1}{4} \cdot \frac{1}{2}\left(\frac{1}{p} + \frac{1}{1-p}\right) = \frac{1}{8} \cdot \frac{1}{p(1-p)} = \frac{5}{8}.$$

The maximum is $\Phi \approx 0.65367$ near $r \approx 0.354$. Combined with (27) this gives

$$M_n - \sqrt{5} \le \frac{0.654\sqrt{5}}{n} + O(n^{-2}),$$

not a clean bound for every $n$: the remainder in (27) may be positive, so the honest statement is that for each $\varepsilon > 0$ the excess is at most $(0.654+\varepsilon)\sqrt{5}/n$ once $n$ is large.

**The two edges.** Near $r = 0$ we have $B(\sigma) \approx 3\sigma^2$, so $\sigma \approx \sqrt{r/3}$ and only the second term of (26) survives:

$$\Phi(r) = \frac{1}{1-p}\sqrt{\frac{r}{3}} + O(r), \qquad r \to 0^+. \tag{28}$$

Near $r = 1$, writing $\tau = 1 - r$, the same expansion at the other end gives $1 - \sigma \approx \sqrt{\tau/3}$ and

$$\Phi(1-\tau) = \frac{1}{p}\sqrt{\frac{\tau}{3}} + O(\tau), \qquad \tau \to 0^+. \tag{29}$$

The two edges are not symmetric. Since $p > 1-p$, the coefficient $\frac{1}{p}$ in (29) is smaller than $\frac{1}{1-p}$ in (28), so the *right* edge, $\{np\}$ just below 1, produces the smaller width. That is the dangerous side.

Combining (27) and (29), when $np$ lies just below an integer,

$$W_n - 1 \;\sim\; \frac{1}{p\sqrt{3}} \cdot \frac{\sqrt{\|np\|_{\mathbb{Z}}}}{n}, \tag{30}$$

which is the square-root law behind the exponent $\frac{3}{2}$: a defect of size $\|np\|_{\mathbb{Z}}$ in the arithmetic costs only $\sqrt{\|np\|_{\mathbb{Z}}}$ in the geometry, because $B$ is tangent to its endpoint values.

## 12. What is and is not established about the exponent

Formula (30) turns the question into one about how well $np$ can be approximated by integers, and that part *can* be settled exactly.

**An exact Diophantine identity.** Let $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2} = -\varphi^{-1}$, and let $L_m = \varphi^m + \psi^m$, $F_m = \frac{\varphi^m - \psi^m}{\sqrt{5}}$ be the Lucas and Fibonacci numbers. Since $p = \frac{1}{2} + \frac{1}{2\sqrt{5}} = \frac{\varphi}{\sqrt{5}}$,

$$L_m p - F_{m+1} = \frac{(\varphi^m + \psi^m)\varphi - (\varphi^{m+1} - \psi^{m+1})}{\sqrt{5}} = \frac{\psi^m(\varphi+\psi)}{\sqrt{5}} = \frac{(-1)^m}{\sqrt{5}\,\varphi^m}, \tag{31}$$

using $\varphi + \psi = 1$. For $m \ge 1$ the right-hand side has modulus below $\frac{1}{2}$, so $F_{m+1}$ is the nearest integer and

$$L_m \, \|L_m p\|_{\mathbb{Z}} = \frac{L_m}{\sqrt{5}\,\varphi^m} = \frac{1 + (-1)^m \varphi^{-2m}}{\sqrt{5}}. \tag{32}$$

For odd $m$ this is *below* $\frac{1}{\sqrt{5}}$ and increases to it; these are the $n = L_m$ with $\{np\} \to 1^-$, the dangerous edge of Section 11. The odd-index Lucas numbers are $4, 11, 29, 76, 199, 521, 1364, 3571, 9349, \ldots$, satisfying $n_{j+1} = 3n_j - n_{j-1}$. Together with Hurwitz's theorem, (32) gives

$$\liminf_{n\to\infty} n\,\|np\|_{\mathbb{Z}} = \frac{1}{\sqrt{5}}, \tag{33}$$

and it shows that the constant $\frac{2}{5}$ of Lemma 4 cannot be raised above $\frac{1}{\sqrt{5}}$.

**The conjectural constant.** Substituting $\|np\|_{\mathbb{Z}} \sim \frac{1}{\sqrt{5}\,n}$ into (30) and granting (27),

$$\liminf_{n\to\infty} n^{3/2}\big(W_n - 1\big) \overset{?}{=} \frac{1}{p\sqrt{3\sqrt{5}}} = 0.5335734771\ldots, \tag{34}$$

$$\liminf_{n\to\infty} n^{3/2}\big(M_n - \sqrt{5}\big) \overset{?}{=} \frac{\sqrt{5}}{p\sqrt{3\sqrt{5}}} = 1.1931065657\ldots$$

The question marks record that (34) inherits both unproved ingredients of Section 11.

**Three statements that must be kept apart.** Let

$$\Lambda_{\text{unif}} = \inf_{n\ge 4} n^{3/2}\big(M_n - \sqrt{5}\big)$$

be the supremum of admissible constants in part (2). Then always

$$\Lambda_{\text{unif}} \le \liminf_{n\to\infty} n^{3/2}\big(M_n - \sqrt{5}\big),$$

and equality is a separate claim: some finite $n$, or some other family of configurations, could give a smaller value. So even a proof of (34) would not by itself identify the best $\lambda$; it would only exhibit an asymptotic obstruction. What Section 9 does prove is $\Lambda_{\text{unif}} \ge \frac{\sqrt{10}}{25} = 0.1264911\ldots$, a factor 9.43 below the conjectured value.

**What would suffice for optimality of the exponent.** To show that $\frac{3}{2}$ cannot be lowered, global optimality of (19) is *not* needed. It is enough to

(a) prove that (22) has a root $w \in (0,1)$ with $v_-(w) < 0 < w < 1 < v_+(w)$, at least for $n = L_m$ with $m$ odd;

(b) prove for that explicit admissible array the bound $\text{width}(x) - \sqrt{5} = O(n^{-3/2})$, using (31) for the size of $\{np\}$.

Then $M_n - \sqrt{5} = O(n^{-3/2})$ along a subsequence, so no universal bound $\text{width}(x) \ge \sqrt{5} + \lambda n^{-\alpha}$ with $\alpha < \frac{3}{2}$ can hold, and part (2) asks for the correct power. Step (a) is a statement about the sign of one explicit scalar function, which is a far smaller task than a global minimisation theorem. Identifying $\Lambda_{\text{unif}}$ or proving (27) would additionally require a genuine moment-space argument: a dual polynomial certificate, or a complete analysis of the optimality conditions ruling out configurations with two interior atoms.

**Why the proof of Sections 6–9 is sharp in order.** Evaluate the chain on the array (19) in the dangerous regime, where $\sigma \to 1$ and $\eta = W_n - 1 \sim (1-\sigma)/(pn)$.

| Quantity | proved bound | value on (19) | loss |
|---|---|---|---|
| $\sum_i d_i$ | $\le 5n\eta$ (12) | $\sim 2(1-\sigma) = 2pn\eta$ | 3.5× |
| $\sum_i d_i^2$ | $\le (\sum_i d_i)^2$ (13) | $\sim (1-\sigma)^2 = p^2n^2\eta^2$ | 4× |
| $\|np\|_{\mathbb{Z}}$ | $\le 125n^2\eta^2$ (16) | $\sim 3(1-\sigma)^2 = 3p^2n^2\eta^2$ | 79.6× |
| $\|np\|_{\mathbb{Z}}$ | $\ge \frac{2}{5n}$ (17) | $\sim \frac{1}{\sqrt{5}\,n}$ along odd Lucas $n$ | 1.12× |

Every entry is tight in order of magnitude; only constants are lost, and after Lemma 4 the arithmetic input is nearly sharp. The exact relation behind the third row is

$$1 - B(\sigma) = (1-\sigma)^2(1+2\sigma),$$

which is asymptotically $3(1-\sigma)^2$ as $\sigma \to 1$; this is (24) again, and the double tangency of $B$ is the reason the exponent is $\frac{3}{2}$ rather than 2. Feeding the sharp constants through the same argument returns

$$\eta \gtrsim \frac{1}{p\sqrt{3\sqrt{5}}}\, n^{-3/2},$$

which is (34). The residual gap between the proved $\frac{\sqrt{10}}{25}$ and the conjectured $\frac{\sqrt{5}}{p\sqrt{3\sqrt{5}}}$ is a factor 9.43, splitting as $\sqrt{79.6} = 8.92$ from the constants in (12), (13), (16) and $\sqrt{1.12} = 1.06$ from Lemma 4.

## 13. Numerics

Solving (20) to 30 digits gives the following. The dangerous subsequence, $n = L_m$ with $m$ odd:

| $n$ | $\{np\}$ | $w_n$ | $\mathrm{width}(u) - 1$ | $n^{3/2}(\mathrm{width}(u) - 1)$ | $n^{3/2}(\mathrm{width}(x) - \sqrt{5})$ |
|---|---|---|---|---|---|
| 4 | 0.894427 | 0.815512 | $8.7707 \cdot 10^{-2}$ | 0.701658 | 1.568954 |
| 11 | 0.959675 | 0.884207 | $1.6911 \cdot 10^{-2}$ | 0.616974 | 1.379596 |
| 29 | 0.984597 | 0.927913 | $3.7103 \cdot 10^{-3}$ | 0.579440 | 1.295668 |
| 76 | 0.994117 | 0.955392 | $8.4485 \cdot 10^{-4}$ | 0.559760 | 1.251662 |
| 199 | 0.997753 | 0.972463 | $1.9553 \cdot 10^{-4}$ | 0.548906 | 1.227392 |
| 521 | 0.999142 | 0.983009 | $4.5637 \cdot 10^{-5}$ | 0.542716 | 1.213550 |
| 1364 | 0.999672 | 0.989514 | $1.0701 \cdot 10^{-5}$ | 0.539094 | 1.205451 |
| 3571 | 0.999875 | 0.993526 | $2.5162 \cdot 10^{-6}$ | 0.536935 | 1.200624 |
| 9349 | 0.999952 | 0.996002 | $5.9254 \cdot 10^{-7}$ | 0.535632 | 1.197709 |
| conjectured limit | 1 | 1 | – | 0.533573 | 1.193107 |

The minimum of $n^{3/2}(\mathrm{width}(u) - 1)$ over $4 \le n \le 10^4$ is 0.5356319, at $n = 9349$; the column decreases monotonically towards the value conjectured in (34). Every entry exceeds the proved bound $\frac{\sqrt{2}}{25} = 0.0565685$ of (18), by a factor between 12 and 9.5.

For ordinary $n$, law (27) is already accurate at moderate size:

| $n$ | $\{np\}$ | $n\big(\mathrm{width}(u) - 1\big)$ | $\Phi(\{np\})$ |
|---|---|---|---|
| 37 | 0.773452 | 0.4501191 | 0.4417257 |
| 100 | 0.360680 | 0.6622113 | 0.6536000 |
| 250 | 0.901699 | 0.2845883 | 0.2839127 |
| 777 | 0.242482 | 0.6331557 | 0.6319304 |
| 2000 | 0.213596 | 0.6180249 | 0.6175447 |
| 5000 | 0.033989 | 0.3316817 | 0.3315599 |

**Status of the numerical evidence.** For $n = 4, 8, 12, 20, 30, 50$ a general constrained minimisation of width over all $n$-good arrays (SLSQP with auxiliary bounds $m \le u_i \le M$, many random starts) returns the value from (20) to seven or eight digits, and no narrower configuration was found for $4 \le n \le 30$. This is local optimisation from random starts, so it is evidence for the conjecture and not a certificate: it rules out nothing globally. Separately, (19) is narrower than the Section 4 patch by a factor of order $n$, for instance $7.48 \cdot 10^{-7}$ against $2.22 \cdot 10^{-3}$ at $n = 15127$, which shows only that the patch is not extremal.

# References


[1] A. Hurwitz, *Über die angenäherte Darstellung der Irrationalzahlen durch rationale Brüche*, Math. Ann. **39** (1891), 279–284. (Used for (33); the special case needed here also follows from the exact identity (31).)

[2] A. Ya. Khinchin, *Continued Fractions*, University of Chicago Press, 1964. (Convergents, equidistribution of $\{np\}$, Section 4.)

[3] M. G. Krein and A. A. Nudelman, *The Markov Moment Problem and Extremal Problems*, Amer. Math. Soc., 1977. (The moment-space framework that a proof of (27) would need.)